Machine Learning for Prediction of Emerging Infectious Diseases

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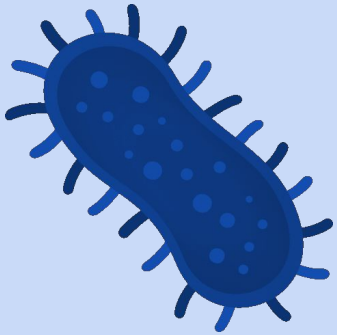
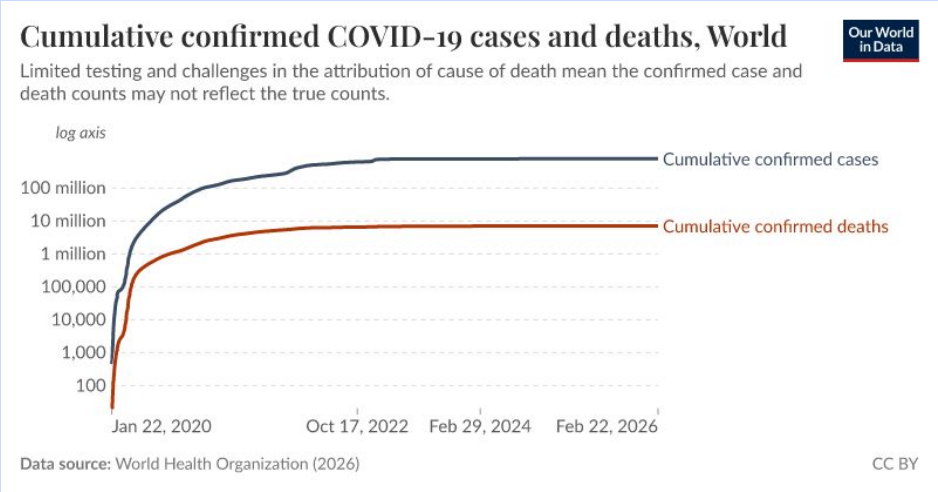


Introduction and Background

- Emerging infectious diseases pose global health threats (SARS/MERS , COVID-19, H1N1)
- Public health informatics enables large-scale data integration
- Machine learning:
  - Analyzes complex, real-time data
  - Identifies early outbreak signals
  - Supports proactive public health response



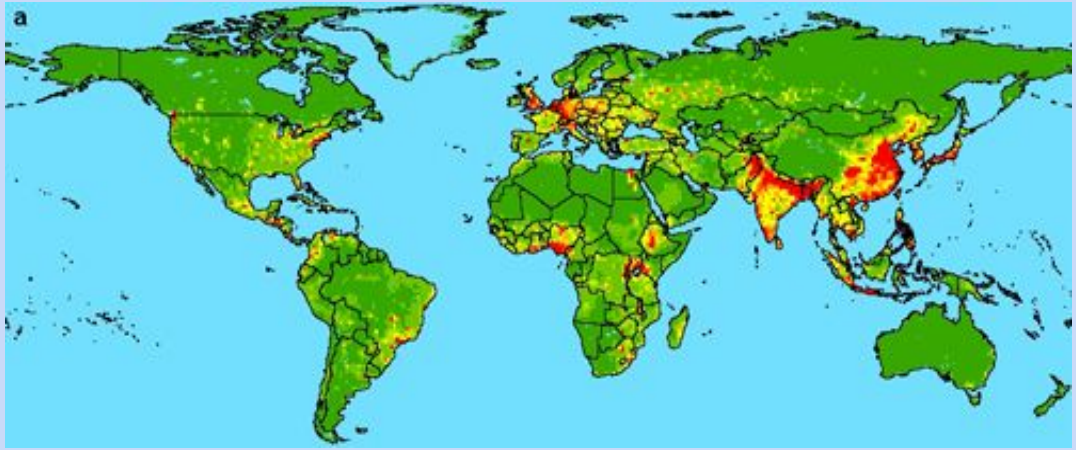
Population/Public Health Impact

- Early detection = early intervention
  - Trigger quarantine measures, vaccinations, travel advisories
- Resource allocation
  - Predict regions most affected
  - Allocate personnel and supplies
- Risk stratification
  - Identify high-risk populations
- Global surveillance
  - ML models can integrate with international datasets
- Reduce healthcare costs, supply chain issues, workforce disruption

Function	Examples
Early warning	• Natural-language processing of news sources to identify outbreaks (Freifeld et al., *JAMIA* 2008) • Unsupervised machine learning of social media data to detect unknown infections (Lim, Tucker, and Kumara, *J Biomed Inform* 2017)
Pathogen classification	• Convolutional neural network model for reading antibiograms (Pascucci et al., *Nat Commun* 2021) • Convolutional neural network model to automate malaria microscopy and diagnosis (Liang et al., *IEEE* 2016)
Risk assessment	• Reinforcement learning of Covid-19 positivity rates to target limited testing in Greece (Bastani et al., *Nature* 2021) • Machine-learning models including random forest and extreme gradient boosting to use syndromic surveillance for Covid-19 risk prediction (Dantas, *PLoS One* 2021)
Source identification	• Automated data mining of electronic medical records to uncover hidden routes of infection transmission (Sundermann et al., *Clin Infect Dis* 2021) • Supervised machine learning in combination with digital signal processing for genomic tracing of Covid-19 (Randhawa et al., *PLoS One* 2020)
Hotspot detection	• Neural computing engine to correlate sound from hospital waiting rooms with influenza spikes (Al Hossain et al., *Proc ACM Interact Mob Wearable Ubiquitous Technol* 2020) • Multilayer perceptron artificial neural network model to detect spatial clustering of tuberculosis (Mollalo et al., *Int J Environ Res Public Health* 2019)
Tracking and forecasting	• Real-time stacking of multiple models to improve forecasts of seasonal influenza (Reich et al., *PLoS Comput Biol* 2019) • Machine learning to combine new data sources for monitoring Covid-19 (Liu et al., *J Med Internet Res* 2020)

Current Challenges

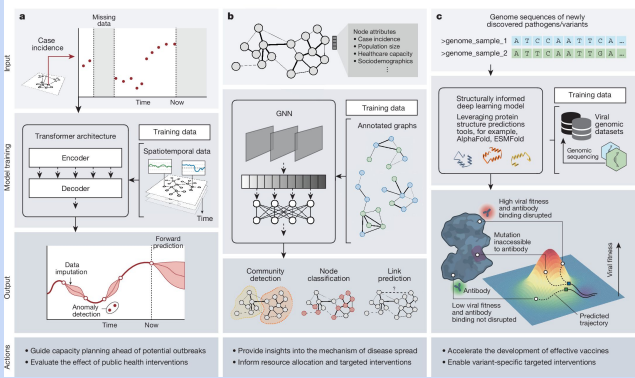
- Obtaining training data
  - Little known about disease mechanisms
  - Emerge in countries with little public health infrastructure
  - Early surveillance data may be biased
- Computing infrastructure
  - Training resource-intensive
- Overfitting training data
  - Limits generalizability
  - Unique disease mechanisms



(Columbia Climate School, 2008)

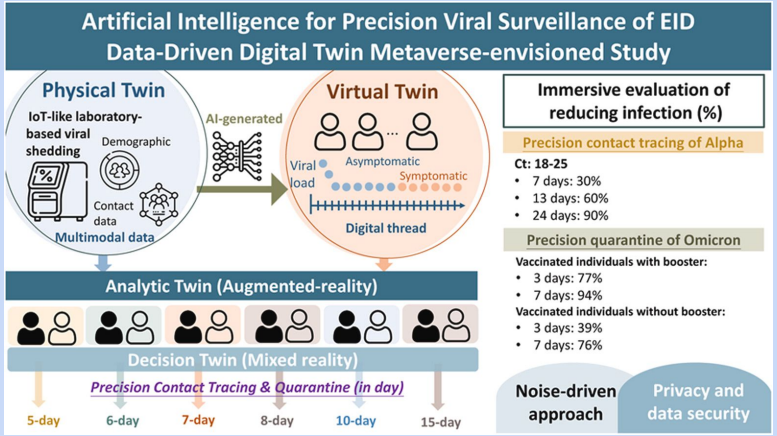
Related Work

- Established directions
  - Widely used for outbreak detection, prediction, and surveillance
  - Applied across the full disease lifecycle, including surveillance, diagnosis, and prognosis
- Modeling approaches used
  - Traditional ML (e.g., regression, ensemble models) and deep learning models
- Integration into surveillance systems
  - AI-enabled systems support real-time outbreak monitoring and response
  - Can detect outbreaks before official reporting
- Capabilities demonstrated
  - Early detection of outbreak signals through pattern recognition
  - Prediction of disease spread and incidence trends
  - Identification of high-risk populations and geographic hotspots



Developing Research

- Developing better models for disease surveillance
- Digital twins
  - Simulate containment measures
  - Integrate real-time data
- Improve public health surveillance in developing countries
  - Can implement to make up for lack of traditional resources
- Monitor public sentiment surrounding epidemic
  - Natural language processing on social media
  - Can adapt public health messaging to situation
- Transfer learning
  - Adapt models based on knowledge from previous epidemics



(Lin et al., 2025)

Conclusion

- Promising for more rapid responses to emerging diseases
- Model limitations must be considered
  - Dependent on training data quality
- Field will grow as AI infrastructure becomes more widespread

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